

Def.

A group  $D_{2n} \stackrel{\text{def}}{=} \langle r, s \mid r^n = s^2 = 1, rs = sr^{-1} \rangle$  is the dihedral group of order  $2n$ .

Prop. Let  $D_{2n}$  be the dihedral of order  $2n$ .

1). For any  $x \in D_{2n} \setminus \langle r \rangle$ ,  $rx = xr^{-1}$ .

2). For any  $x \in D_{2n} \setminus \langle r \rangle$ ,  $|x| = 2$ .

3).  $D_{2n} = \langle s, sr \rangle$ .

4). If  $n = 2k$  is even with  $n \geq 4$ , then  $r^k$  has order 2 and  $\langle r^k \rangle = \{1, r^k\} = Z(D_{2n})$ .

5). If  $n$  is odd with  $n \geq 3$ , then  $Z(G) = \{1\}$ .

Proof. Let  $x \in D_{2n}$  be given. Then  $x = r^a s$  for some  $a = 0, 1, \dots, n-1$ , by  $x \notin \langle r \rangle$ .

So,  $rx = r^{a+1}s = r^a \cdot r \cdot s = r^a \cdot sr^{-1} = xr^{-1}$ .

And,  $x^2 = r^a s r^a s = r^{a-1} s r^{-1} r^a s = \dots = s r^{-a} \cdot r^a \cdot s = s^2 = 1$ , and  $x \neq 1$ ,

so  $|x| = 2$ . The 3) is given by  $\langle s, sr \rangle \subseteq D_{2n}$  and  $ssr = r \in \langle s, sr \rangle$ , so  $\langle s, sr \rangle \supseteq \langle s, r \rangle = D_{2n}$ .

To determine the Center of  $D_{2n}$ , observe that: given  $a \in \{0, 1, \dots, n-1\}$ ,

$r^a \in Z(D_{2n}) \Leftrightarrow r^a s = s r^a = s r^{-a} \Leftrightarrow r^{2a} = 1$ , by the cancellation law.

Since  $r^{2a} = 1$  if and only if  $n \mid 2a$ , so  $r^a \in Z(D_{2n})$  if and only if

$$a = \begin{cases} 0 & \text{if } n \text{ is odd} \\ 0, \frac{n}{2} & \text{if } n \text{ is even.} \end{cases}$$

And, given  $x \in D_{2n} \setminus \langle r \rangle$ ,  $x \in Z(D_{2n})$  implies that  $xr = rx = xr^{-1} \Rightarrow r^2 = 1$ .

But, it contradicts to  $|r| = n \geq 3$ . Consequently, for  $n \geq 3$ ,

$$Z(G) = \begin{cases} \{1\} & \text{if } n \text{ is odd} \\ \{1, r^k\} & \text{if } n = 2k \text{ is even} \end{cases}$$

□

Prop. If  $n$  is even, then  $D_{2n} \cong \langle (1\ 2 \dots n), (2\ n)(3\ n-1) \dots (\frac{n}{2}\ \frac{n}{2}+2) \rangle \leq S_n$   
 If  $n$  is odd, then  $D_{2n} \cong \langle (1\ 2 \dots n), (2\ n)(3\ n-1) \dots (\frac{n+1}{2}\ \frac{n+1}{2}+1) \rangle \leq S_n$ .

Proof In case of  $n$  is even number, observe that

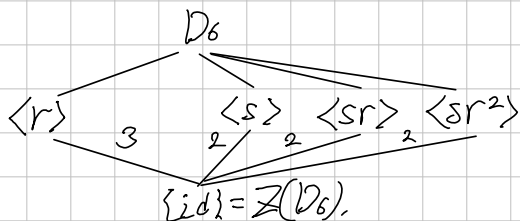
$$(1\ 2 \dots n)(2\ n)(3\ n-1) \dots (\frac{n}{2}\ \frac{n}{2}+2)(1\ 2 \dots n) \\ = (1)(2\ n)(3\ n-1) \dots (\frac{n}{2}\ \frac{n}{2}+2)$$

Moreover,  $(1\ 2 \dots n)^n = \text{id}$  is clear, and since  $(2\ n) \dots (\frac{n}{2}\ \frac{n}{2}+2)$  is product of disjoint two-cycles, so it has order 2.

So, corresponding to  $r = (1\ 2 \dots n)$  and  $s = (2\ n) \dots (\frac{n}{2}\ \frac{n}{2}+2)$ , it satisfies the representation of the definition. Similarly,  $n$  is odd, can show.  $\square$

$D_6$ .

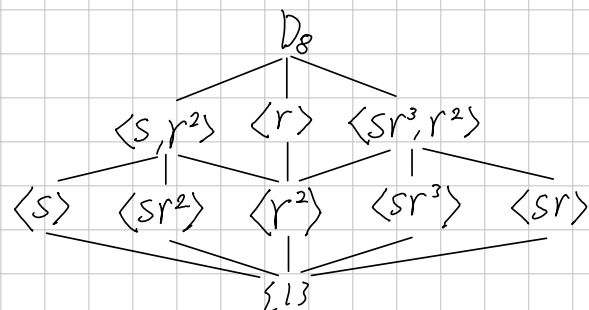
By the theorem above,  $D_6 \cong S_3$ . Now, explicitly,



Normalizer:

$$N_{D_6}(\langle r \rangle) = D_6$$

$$\sqrt{D_8} \\ D_8 = \langle r, s \mid r^4 = s^2 = 1, rs = sr^{-1} \rangle$$



Normalizer: Since index of 2 subgroups must be normal,  
the normalizer of  $\langle s, r^2 \rangle$ ,  $\langle r \rangle$ ,  $\langle sr^3, r^2 \rangle$  is just  $D_8$ .  
 $N_{D_8}(\langle s \rangle) = \langle s, r^2 \rangle$ , because: since  $|\langle s, r^2 \rangle : \langle s \rangle| = 2$ ,  $\langle s, r^2 \rangle \leq N_{D_8}(\langle s \rangle)$ , but  
 $r\langle s \rangle r^{-1} = \{1, sr^2\} \neq \langle s \rangle$ , so  $N_{D_8}(\langle s \rangle) \neq D_8$ .

Similarly  $N_{D_8}(\langle sr^2 \rangle) = \langle s, r^2 \rangle$ . But,  $N_{D_8}(\langle r^2 \rangle) = D_8$ , because:  
first way: since  $|\langle s, r^2 \rangle : \langle r^2 \rangle| = |\langle r \rangle : \langle r^2 \rangle| = 2$ , therefore

$$\langle s, r^2 \rangle, \langle r \rangle \subseteq N_{D_8}(\langle r^2 \rangle), \text{ so } N_{D_8}(\langle r^2 \rangle) = D_8.$$

Alternatively, since  $sr^2s^{-1} = sr^2s = r^{-2}s \cdot s = r^2$ , so  $s, r \in N_{D_8}(\langle r^2 \rangle)$ ,  
so  $\langle s, r \rangle \subseteq N_{D_8}(\langle r^2 \rangle)$ . ( $r \in D_8$  clearly normalizes  $\langle r^2 \rangle$ ).

Centralizer: First,  $C_{D_8}(D_8) = \{1, r^2\} = \langle r^2 \rangle$  because:  $sr^2 = r^{-2}s$  and  
clearly  $r^2$  commutes all elements in  $\langle r \rangle$ , so  $r^2$  commutes all elements of  $D_8$ .  
but,  $sr^3 = rs \neq r^3s$ , so  $r, s \notin Z(G)$  and  $sr^2 \notin Z(G)$  because  
 $sr^2 \in Z(G) \Rightarrow sr^2r^{-2} = s \in Z(G)$  since the center is a subgroup.

Moreover,  $sr \cdot s = r^3ss = r^3 \neq ssr = r$ , so  $sr \notin Z(G)$ . Thus  $Z(G) = \langle r^2 \rangle$

Next,  $C_{D_8}(\langle r \rangle) = \langle r \rangle$ , because: since  $rs = sr^3 \neq sr$ , so  $s \notin C_{D_8}(\langle r \rangle)$  and  
clearly all elements in  $\langle r \rangle$  commute with the all elements in  $\langle r \rangle$ , so  
 $\langle r \rangle \subseteq C_{D_8}(\langle r \rangle) \subsetneq D_8$ , so the Lagrange's theorem gives the result.

Commutator: Since the derived set is actually normal subgroup and  
it induces the largest abelian quotient, so  $D_8' = \langle r^2 \rangle$ , because  $D_8 / \langle r^2 \rangle \cong \mathbb{Z}_2 \times \mathbb{Z}_2$ ,  
because it contains two elements order 2, but it is not cyclic.

$\mathbb{D}_{10}$